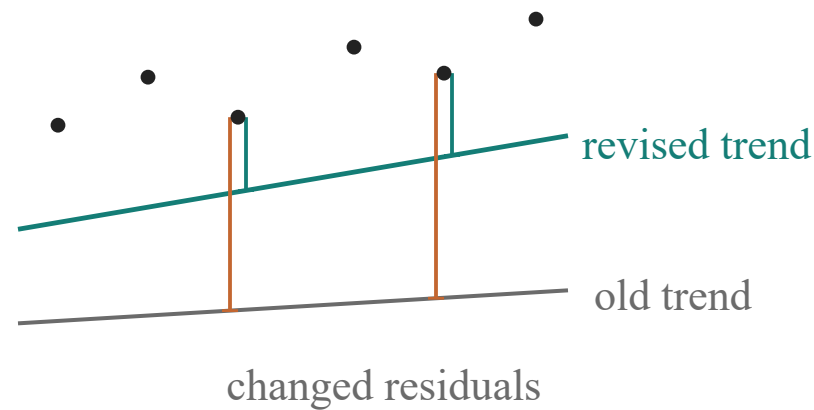


(a) Trend revision

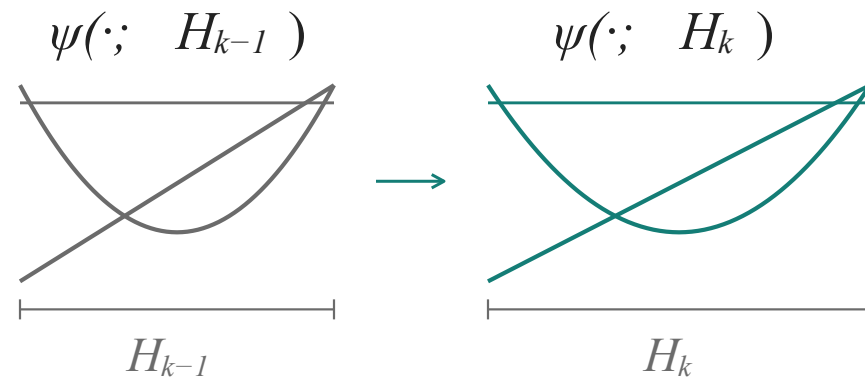


$$\Delta h_u = - \textcolor{teal}{R}_{u\beta} \Delta \beta$$

$\textcolor{teal}{R}_{u\beta}$ retained cross statistic

residual-only summary becomes stale

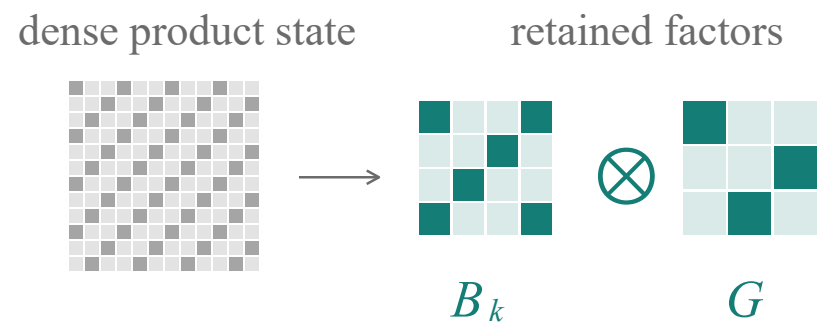
(b) Changing coordinates



$$(R, r) \rightarrow (\textcolor{teal}{M}^T R \textcolor{teal}{M}, \textcolor{teal}{M}^T r)$$

likelihood evidence,
not latent-state identity

(c) Structured recovery



$$R_{uu,k} = B_k \otimes G$$

Schur–Sylvester solve

$$\downarrow$$

$$q_k(\beta, u_k)$$

$O(M^3)$ recovery $O(M^2)$ persistent state